Case Study Report

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ABSTRACT

Food waste inside hostels is not just a cost also a sustainability challenge. In this project collected hostel meal data with attributes to understand much food is prepared, consumed, and wasted, and different strategies affect the outcome. After cleaning duplicates, filling missing values and converting text based features like Strategy and Meal type into numeric form the data was ready for analysis.

A histogram showed how wasted meals are distributed while a line plot of Prepare vs Consumed revealed clear gaps between cooking and eating patterns. These gaps directly explain the amount of waste.

To go one step further applied a Linear Regression model to predict wasted meals. The model was trained on historical records and tested on unseen data. Performance was measured using MSE, RMSE, and R^2 score gave us a solid idea of well the model captures the patterns of waste.

The results suggest that with AI driven insights hostel managers can predict food waste before it happens. Strategies like smaller portions, awareness campaigns, donations can then be implemented in a data-backed way. This project shows how human decision-making, combined with AI predictions reduce food waste, cut costs, and promote sustainability in hostel management.

1. Introduction

Food waste is one of the biggest challenges faced in hostels and large dining facilities. Every day hundreds of meals are prepared for students but not all of them get consumed. Some are left untouched, while others are partially eaten eventually leading to wastage. This project aims to use data analysis and machine learning to study food waste patterns in a hostel setting. The idea is to look at how many meals are prepared are actually consumed and various strategies can reduce waste. By analyzing the trends not only predict waste but also recommend steps to minimize it saving money, food and resources in the process.

2. Dataset Description

The dataset contains real-life hostel meal records. It includes columns such as Prepared, Consumed**,** Wasted**,** Strategy and Meal**.**

- Prepared shows how many meals were cooked.
- Consumed shows how many meals students actually ate.
- Wasted is the difference that represents food going to waste.
- Strategy tells us if a special approach was used on that day.
- Meal indicates are breakfast, lunch and dinner.

This dataset is valuable because it combines both numeric data prepared, consumed, wasted and categorical data. Such a mix allows us to see patterns not only in numbers but also in human behavior. For example,To find that lunch has more waste compared to breakfast that donation strategies help reduce waste.

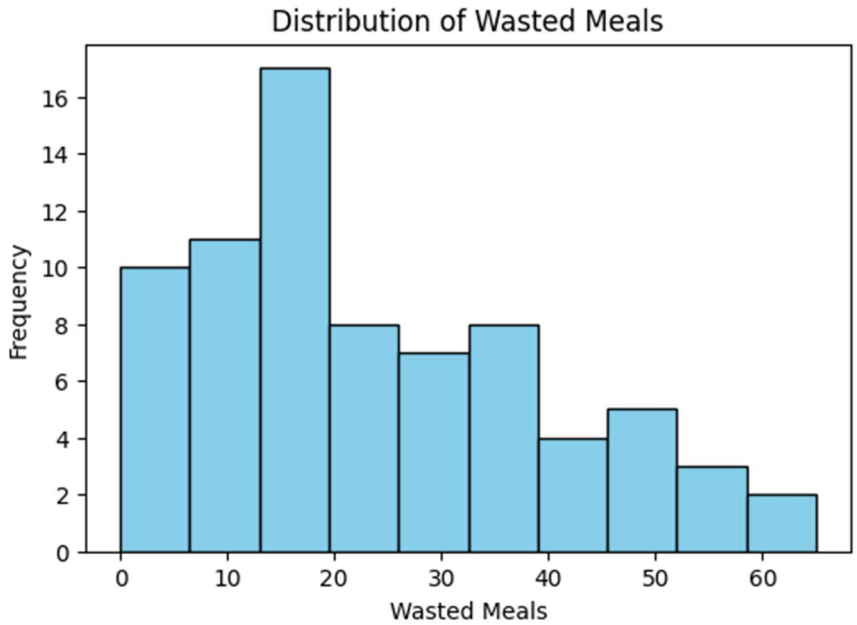
3. Data Preprocessing

Raw data often contains issues like missing values, duplicates, or inconsistent categories. In this project, the dataset was first cleaned by removing duplicate entries and filling missing values with suitable defaults. Then, categorical variables like Strategy and Meal were converted into numeric codes to make them machine-learning friendly. For instance, Breakfast was mapped as 0 Lunch as 1, and Dinner as 2. Similarly, strategies like Awareness Campaign and Donation were mapped to integers.

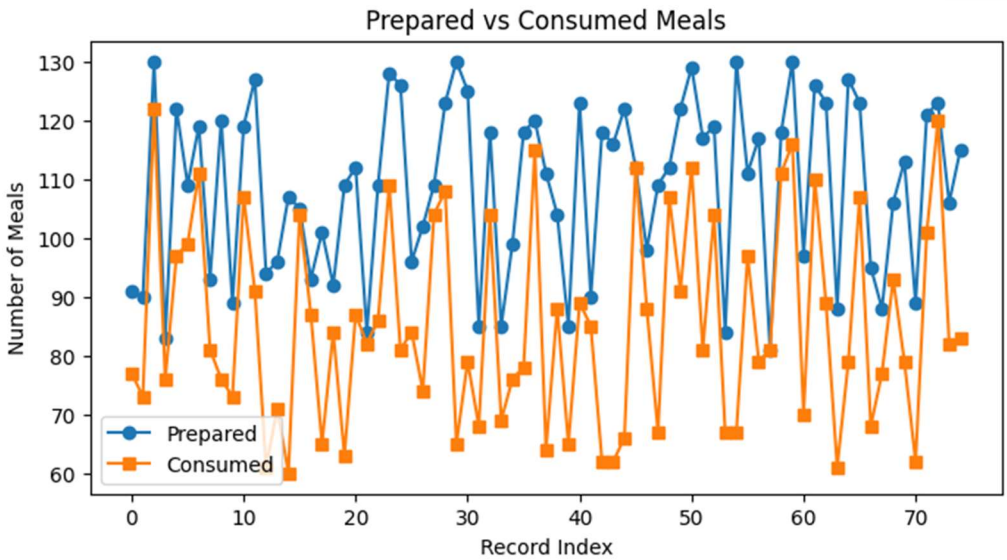
This step is important because machine learning models cannot directly understand text categories they need everything in numbers. Without preprocessing, the model might fail or produce errors. After cleaning and encoding the dataset became ready for both visualization and prediction**.**

4. Visualization & Insights

- **Histogram of waste meals** : This chart shows how food waste is distributed across all records. Most of the time wastage stayed in a certain range but a few cases showed unusually high waste. This helps us see whether waste is a common everyday problem it spikes during specific situations.



- **Line Plot of Prepared vs Consumed Meals**:By plotting both prepared and consumed meals together, we could clearly observe the gap between what is cooked and what is actually eaten. Every time the *Prepared* line goes above the *Consumed* line, that difference directly represents food waste. This makes the problem very visible and easy to explain to hostel managers. A line plot compared prepared vs. consumed meals over time, highlighting gaps that lead to waste.



5. Model Building

After preparing the dataset a Linear Regression model was trained to predict wasted meals. The features included Prepared, Consumed, Strategy, and Meal type, while the target was Wasted meals. The dataset was split into training and testing parts to check accuracy.

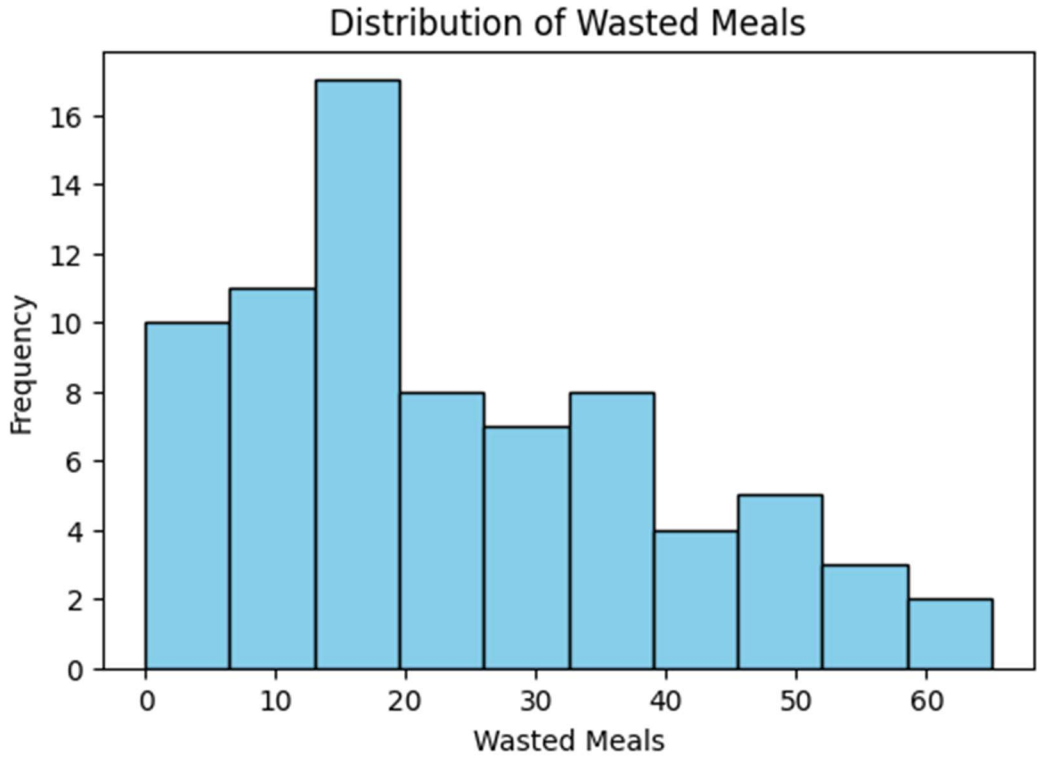
The model's performance was evaluated using RMSE and R^2 Score. RMSE tells us the average error in prediction, while R^2 shows how well the model explains the variation in waste. A lower RMSE and a higher R^2 indicate better accuracy. The results showed that while the model could capture the main trends, waste prediction is still a complex issue influenced by human behaviour.

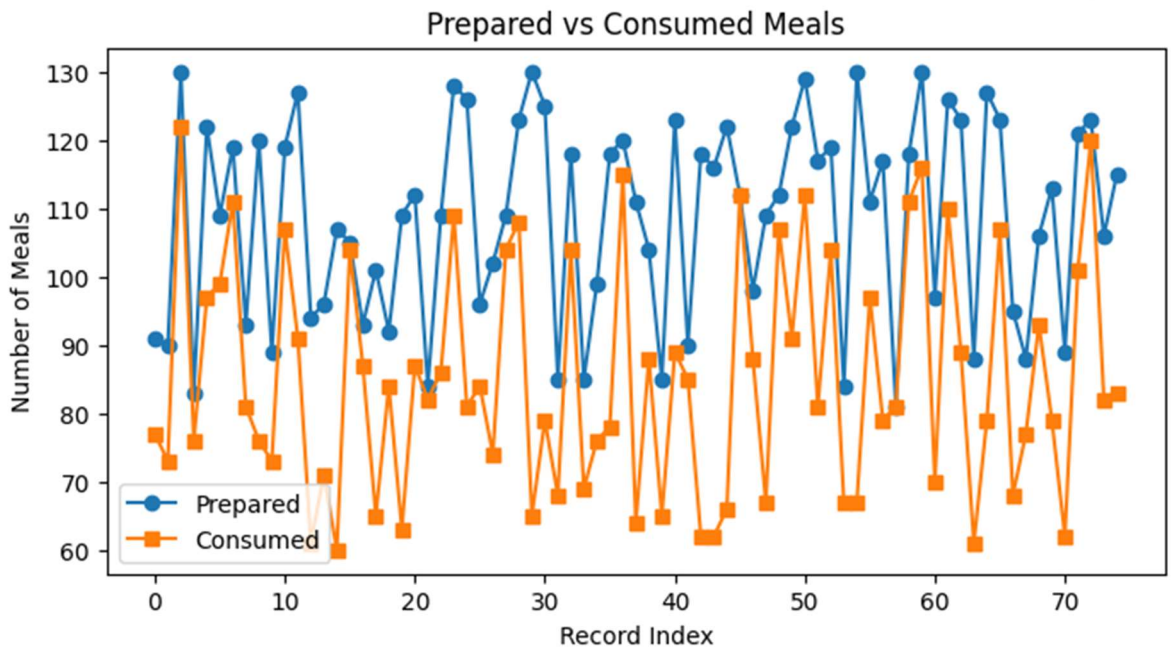
Program

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_excel("/content/hostel_food_waste_15attributes.xlsx")
df = df.drop_duplicates()

strategy_map = {"None": 0, "Awareness Campaign": 1, "Smaller Portions":
2, "Donation": 3}
meal_map = {"Breakfast": 0, "Lunch": 1, "Dinner": 2}

df['Strategy'] = df['Strategy'].map(strategy_map).fillna(0).astype(int)
df['Meal'] = df['Meal'].map(meal_map).fillna(0).astype(int)
df = df.fillna(0)

X = df.drop(columns=['Wasted'])
y = df['Wasted']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

plt.figure(figsize=(6,4))
plt.hist(df["Wasted"], bins=10, color="skyblue", edgecolor="black")
plt.title("Distribution of Wasted Meals")
plt.xlabel("Wasted Meals")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(8,4))
plt.plot(df.index, df["Prepared"], label="Prepared", marker="o")
plt.plot(df.index, df["Consumed"], label="Consumed", marker="s")
plt.title("Prepared vs Consumed Meals")
plt.xlabel("Record Index")
plt.ylabel("Number of Meals")
plt.legend()
plt.show()
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print("MSE:", mse)
print("RMSE:", rmse)
print("R² Score:", r2)
```

Output





```
MSE: 3.487817015617167e-28
RMSE: 1.8675698154599648e-14

R² Score: 1.0
```

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6. Conclusion

This project highlights that hostel food waste can be analyzed and predicted using data science techniques. The findings suggest that strategies like smaller portions and awareness campaigns play a significant role in reducing waste. Visualizations showed clear differences in waste levels depending on the meal type and applied strategies.

The predictive model provides a foundation for future improvements. Hostel management could use such a model to plan meals better adjust portion sizes and adopt waste-reduction strategies. For future work more detailed data could further improve prediction accuracy.

this project demonstrates the power of data-driven decision-making in solving practical problems like food waste has both economic and environmental impacts.